researchers are implementing end-to-end encryption, blockchain-based authentication, and quantum-secure communication protocols. Such systems aim to ensure data confidentiality and operational resilience against electronic warfare and cyberattacks.

7.3 Navigation and Autonomy Challenges

7.3.1 Precision in GPS-denied environments

One of the most significant navigation challenges in modern drone operations is maintaining positional accuracy and flight stability in GPS-denied environments. Global positioning system (GPS) signals are the cornerstone of drone navigation, enabling precise localization, route planning, and autonomous control. However, in many operational contexts – such as indoor facilities, dense urban areas, mountainous terrains, forests, or military zones – GPS signals become weak, obstructed, or deliberately jammed, leading to loss of positional awareness and mission failure.

To address this limitation, researchers have developed a range of alternative navigation technologies that compensate for GPS unavailability. Inertial navigation systems (INS), which rely on accelerometers and gyroscopes, provide short-term positional estimates based on motion tracking, though they tend to accumulate errors over time. Visual odometry (VO) and simultaneous localization and mapping (SLAM) techniques have gained prominence by using onboard cameras and LiDAR sensors to build real-time maps of the environment and estimate the drone’s location relative to recognizable features. These methods allow drones to navigate autonomously through vision-based perception rather than relying solely on satellite signals.

Moreover, sensor fusion, the integration of multiple data sources such as barometric pressure sensors, magnetometers, infrared depth sensors, and